

An Early Warning System for Liquidity Stress in Cryptocurrency Markets Using Trade Flow Analysis and Machine Learning

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Abstract

Most research on liquidity crises relies on expensive institutional data or coarse daily aggregates — reconstructing why crashes occurred rather than detecting them as they unfold. We present an early warning system for liquidity stress in Binance spot cryptocurrency markets using freely available high-frequency trade data, deployable against the public Binance API with no proprietary feeds required. Our contributions are fourfold: (1) a natively stationary microstructure feature set (OFI, RV, Kyle’s λ , ILLIQ, VWAP deviation, intensity, and TCI) derived from publicly available trade data; (2) a purged expanding walk-forward framework preventing temporal leakage; (3) a comparative study of five ML architectures under a unified protocol across binary and multiclass formulations, five seeds, and pooled/asset-specific configurations; and (4) a three-tier external validation framework addressing the self-referentiality of HMM-labelled pipelines through a controlled global-vs-local HMM experiment with genuinely independent validators.

XGBoost achieves a binary weighted F1 of **0.9706** on the final out-of-sample window (18.8M training rows) with near-zero seed variance (range: 0.0006), substantially outperforming a persistence baseline (F1-W 0.7891) and confirming genuine predictive skill beyond regime momentum. Against external reference definitions, the framework provides 56 minutes of advance warning before the documented FTX trigger and 108 minutes before the Terra-Luna structural break. The global HMM is near-deterministic (2.66×10^{-7} maximum parameter difference across independent runs), while locally-fitted HMMs produce stress percentages of 6.0% to 94.3% for identical periods—demonstrating systematic miscalibration under narrow training horizons. Against three independent external validators, the global HMM achieves substantial to almost perfect Cohen’s κ agreement, while local variants collapse across multiple folds. A cross-evaluation falsification test confirms XGBoost is not recovering HMM label geometry: stress F1 drops from 0.988 to 0.480 when a locally-trained model is evaluated against global labels. Finally, we provide a live inference system deployable against the public Binance API, bridging the gap between academic modelling and real-world trading infrastructure.

*Lead author and corresponding author. G. Kalu conceived the technical research design and implemented the complete data pipeline, microstructure feature engineering, experimental framework, model training, explainability analysis, external validation framework, live inference system, and manuscript preparation.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Economics*.

Keywords

cryptocurrency, market microstructure, liquidity stress, machine learning, HMM regime labelling, early warning systems, walk-forward validation

*In memory of Adanna Chisom Kalu (2000–2025).
I miss you — your big brother.*

1 Introduction

The rapid growth of cryptocurrency markets since the launch of Bitcoin in 2009 has introduced financial risks characterised by extreme volatility, continuous 24-hour trading, and the absence of traditional circuit breakers, creating a perfect storm of liquidity vulnerabilities. The collapse of Terra-Luna in May 2022, the FTX bankruptcy (November 2022), and the COVID-induced crash (March 2020) demonstrated that liquidity stress can materialise within minutes in crypto markets, wiping out positions before conventional risk systems can respond.

Unlike equity markets, where decades of tough lessons have led to market makers and regulatory halts providing structural buffers, cryptocurrency markets largely operate without these stabilising mechanisms. While regulation has improved, the relatively unfettered nature of crypto exchanges makes early detection of stress regimes operationally critical.

Most existing work analyses stress events in hindsight using coarse daily or hourly data [5, 7], relies on expensive proprietary order book feeds [10], or focuses on short-term price forecasting rather than structural regime classification [17]. Studies addressing high-frequency stress detection typically use static held-out test sets, ignoring temporal dependencies and data leakage risks [12].

A deeper concern—which became apparent during the operational development of our pipeline—is self-referentiality. When stress regimes are defined by an unsupervised model fitted on the exact same data window used for classification, the resulting framework risks circularity.

This paper addresses these gaps. Specifically, our contributions are:

- (1) We construct a natively stationary microstructure feature set from publicly available Binance trade data, encompassing OFI, RV, Kyle’s λ , ILLIQ, VWAP deviation, trade intensity, and TCI.
- (2) We design a purged expanding walk-forward validation framework with a 30-minute embargo to strictly prevent temporal leakage.
- (3) We benchmark five ML architectures (Logistic Regression, Random Forest, XGBoost, LSTM, CNN-GAF) across binary and multiclass formulations to establish the optimal inference engine.
- (4) We address label self-referentiality through a three-tier external validation framework, demonstrating the necessity of global HMM calibration over local fitting against genuinely independent validators—price drawdowns, documented crisis timestamps, and cross-asset contagion.

To ensure reproducibility and bridge the gap to real-world trading infrastructure, the complete pipeline, trained models, and a live API inference system are available at <https://github.com/GoodieGoody/maic>.

2 Related Work

Market microstructure theory provides the theoretical foundation for our feature set. Roll (1984) [20], Glosten and Milgrom (1985) [11], and Kyle (1985) [14] established the core frictions of financial markets—transaction costs, inventory risk, and adverse selection. Amihud (2002) [2] demonstrated that illiquidity commands a return premium, bridging microstructure to asset pricing.

A key enabling result is Lee and Ready (1991) [16], who developed a method to classify trades as buyer- or seller-initiated from price data alone—the exact mechanism that allows our system to compute OFI from freely available Binance trade data without requiring expensive proprietary order book feeds. Chordia et al. (2002) [8] confirmed that signed order flow is a significant determinant of liquidity conditions, with imbalances in either direction reducing liquidity through market-maker inventory effects.

Brunnermeier and Pedersen (2009) [7] formalised liquidity spirals: traders selling under stress push prices down, triggering margin calls and further selling in a self-reinforcing cascade. Kirilenko et al. (2014) [13] showed this mechanism visible in basic trade data during the 2010 Flash Crash, which erased more than \$1 trillion in market value within minutes before nearly fully recovering [?]. The Terra-Luna collapse [6] and COVID-19 cross-asset contagion [1] confirm these spirals operate in crypto. Our system is specifically designed to detect the earliest microstructure rumblings of this doom loop—before it becomes self-reinforcing.

Kercheval and Zhang (2015) [12] established that ML models predict short-term price movements from trade features. Sirignano and Cont (2018) [21] proved that pooling data across assets outperforms asset-specific models — directly motivating our pooled multi-asset configuration. López de Prado (2018) [18] formalised purged walk-forward cross-validation to prevent temporal leakage, which inflates reported performance in naive splits. SHAP [19] provides the regulatory-grade explainability financial risk systems require.

Table 1: Sampling Windows and Market Context

W	Period	Context	Key Event
0	Feb–Jul 2020	COVID crash	COVID-19 (Mar 2020)
1	Nov 2020–May 2021	Bull run	May 2021 crash
2	Nov 2021–May 2022	Peak + contagion	Terra-Luna
3	Nov 2022–Apr 2023	Post-FTX recovery	FTX bankruptcy
4	Jul–Dec 2024	Post-halving	—

Every study above reconstructs a crisis retrospectively, after the underlying data is already complete. This paper instead evaluates the framework prospectively: it issues warnings 56–108 minutes ahead of documented crisis onsets, measured against externally verified reference definitions rather than the model’s own retrospective labels.

3 Data and Methodology

3.1 Data Collection

Raw trade data was collected from Binance for BTCUSDT, ETHUSDT, and SOLUSDT across five deliberate six-month windows capturing distinct market regimes (Table 1). Window 0 (Feb–Jul 2020) is included in all fold training sets to provide exposure to the COVID-19 crash—the only global macro-driven stress event in the dataset. SOLUSDT begins November 2020 (Binance listing date) and is absent from Window 0. Each trade record includes timestamp, price, quantity, and buyer-initiated flag. Trades were aggregated to 300-second bars (≈ 2.1 million observations per asset). Prior to feature engineering, a three-layer quality audit verified schema consistency, flagged duplicate trade IDs, and removed extreme price outliers; zero-volume bars were forward-filled to ensure feature robustness.

3.2 Microstructure Feature Engineering

Seven features are computed at the 300-second aggregation frequency, each grounded in market microstructure theory. **OFI** measures directional pressure of informed trading: $OFI_t = (\sum_{\text{buy}} q_i - \sum_{\text{sell}} q_j) / \sum q_k$, bounded in $[-1, 1]$. **RV** captures model-free price instability as the sum of squared intraday returns $RV_t = \sum r_{i,t}^2$ [3]. **Kyle’s λ** estimates price impact via OLS of price changes on signed order flow. **ILLIQ** adapts Amihud’s measure to 300s: $|r_t|/V_t$. **VWAP deviation**, **trade intensity** (normalised count), and **Trade Concentration Index (TCI)** (Herfindahl-style concentration of trade sizes) complete the feature set. All seven are natively stationary — each measures a rate, ratio, or local dynamic rather than a price level.

Only the raw price series is non-stationary. Fractional differencing [18] is applied with asset-specific minimum differencing orders: BTCUSDT $d^* = 0.3$, ETHUSDT $d^* = 0.4$, SOLUSDT $d^* = 0.2$.

3.3 Regime Labelling

Stress labels are generated by a 3-state Gaussian HMM fitted per asset on $\{RV_{300s}, OFI_{300s}, \hat{\lambda}_{300s}, \text{intensity}_{300s}\}$ using the Baum-Welch EM algorithm [4] as implemented in `hmmlearn` (diagonal covariance, 100 EM iterations, 5 seeds, best log-likelihood retained). States map to calm/elevated/stress by RV mean rank. Binary classification collapses calm and elevated into non-stress.

Decoding procedure. State sequence decoding uses the Viterbi algorithm [22] (`GaussianHMM.predict()`), which finds the globally most probable state sequence given the full observation sequence. Applied to the complete dataset, this means early-sample states are assigned with knowledge of future observations—a form of smoothing that is standard practice in retrospective regime labelling but constitutes look-ahead relative to a real-time system. We adopt global Viterbi deliberately: the alternative—local Viterbi restricted to fold-available data at each boundary—produces catastrophically miscalibrated labels (stress rates of 6.0% to 94.3% for identical assets and periods, as demonstrated in Section 4.4). The global solution is near-deterministic (2.66×10^{-7} maximum parameter difference across independent runs), confirming it represents the unique convergent solution at this data scale rather than a random local optimum. The three-tier external validation framework in Section 3.4 provides the empirical justification for this design choice.

3.4 External Validation Design

A fundamental concern in HMM-labelled pipelines is self-referentiality. We address this through two analyses. First, a **stability proof**: comparing April 2026 backup model parameters against current production models confirms the global HMM is near-deterministic (ETH/SOL: 0.00×10^0 ; BTC: 2.66×10^{-7} maximum parameter difference). Second, a **controlled experiment**: local HMMs fitted using only training-available data at each fold boundary are compared against the global HMM using three genuinely independent validators—none of which are HMM training features:

- (1) **Price drawdown**—the HMM never observed raw price levels
- (2) **Crisis timestamps**—externally documented in the literature
- (3) **Cross-asset simultaneous stress**—HMM fitted per-asset independently, never saw joint dynamics

Agreement is quantified via Cohen’s κ [9], interpreted per Landis & Koch [15]. RV, OFI, Kyle’s λ , and intensity are explicitly excluded as validators to prevent circularity.

3.5 Lead-Time Methodology

Predictive lead time is the interval between the model’s first sustained signal ($P > 0.85$ for ≥ 2 consecutive 300s bars) and the reference onset. To avoid self-referentiality, lead time is measured against two independent reference definitions in addition to HMM-defined stress onset: (i) externally documented crisis timestamps from the academic literature and contemporaneous news record, and (ii) the first breach of a 5% rolling price drawdown from the preceding six-hour peak—a price-derived measure the HMM never observed. The continuity condition (≥ 2 consecutive bars) filters spurious spikes while preserving sensitivity to structural breaks.

3.6 Experimental Design

Purged expanding walk-forward cross-validation with four folds and a 30-minute embargo. Models trained in asset-specific and pooled configurations across binary and multiclass formulations, five seeds. The input matrix $\mathbf{X}$ consists of the seven microstructure features plus the fractionally differenced price series (28 features for pooled models with asset identifier, 27 for asset-specific). Five

Table 2: Production Stability—Mean $\pm$ Std, 5 Seeds, All Folds

Mode	Model	Acc.	F1-W	F1-M	F1-S
Multi.	LR	.658 $\pm$.132	.634 $\pm$.154	.517 $\pm$.104	.546 $\pm$.189
	RF	.878 $\pm$.084	.888 $\pm$.080	.811 $\pm$.115	.875 $\pm$.056
	XGB	.905 $\pm$.067	.912 $\pm$.064	.852 $\pm$.094	.885 $\pm$.061
	LSTM	.666 $\pm$.151	.661 $\pm$.156	.502 $\pm$.036	.582 $\pm$.076
	CNN-GAF	.417 $\pm$.083	.415 $\pm$.095	.292 $\pm$.041	.198 $\pm$.118
Binary	LR	.779 $\pm$.070	.784 $\pm$.089	.703 $\pm$.065	.559 $\pm$.096
	RF	.945 $\pm$.040	.945 $\pm$.040	.920 $\pm$.035	.881 $\pm$.046
	XGB	.946 $\pm$.054	.947 $\pm$.052	.925 $\pm$.051	.889 $\pm$.064
	LSTM	.817 $\pm$.072	.822 $\pm$.072	.731 $\pm$.031	.600 $\pm$.101
	CNN-GAF	.541 $\pm$.162	.541 $\pm$.140	.424 $\pm$.098	.209 $\pm$.156

F1-W=Weighted, F1-M=Macro, F1-S=Stress class

architectures: Logistic Regression (ℓ_2), Random Forest (cuML, 200 trees), XGBoost (GPU hist, 300 estimators), LSTM (2-layer, 128 units, BF16), and CNN-GAF (Gramian Angular Field, 4-layer CNN).

3.7 Live Inference System

The production XGBoost model is deployed via `scripts/12_inference.py`, fetching live aggregated trades from the Binance public REST API (GET `/api/v3/aggTrades`) over a strict 300-second time window matching the training bar construction exactly. A mandatory pagination loop addresses a critical operational constraint: during high-volatility events trade intensity routinely exceeds the exchange’s 1,000-record per-call limit, and a single call would silently truncate the feature window. The system applies fractional differencing to price in-place before inference, using a relaxed weight-truncation threshold suited to the live history window available (training uses years of history; live inference has on the order of minutes), and declares a WARNING when $P(\text{stress}) > 0.85$ for ≥ 2 consecutive bars — the continuity condition that distinguishes genuine regime deterioration from transient microstructure noise.

4 Results and Discussion

4.1 Headline Results

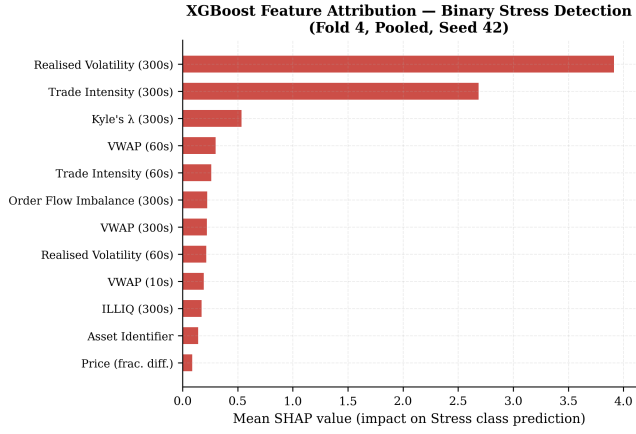
Table 2 reports stability across five seeds and four expanding walk-forward folds (pooled, fractionally differenced).

XGBoost dominates across all metrics. At Fold 4 (18.8M training rows) it achieves binary F1-W of **0.9706** with seed range 0.0006 — near-deterministic at this data scale. RF is a strong second (0.9452 ± 0.0398). Neural architectures substantially underperform: domain-informed microstructure features aggregate temporal dynamics into single-timestep representations, making gradient boosting more effective than sequential architectures that would need to re-learn what feature engineering already encodes. The pooled model matches or exceeds asset-specific performance across all three assets, validating the universal price formation hypothesis [21]. Specifically, pooling exposes the model to cross-asset stress contagion dynamics — the simultaneous deterioration patterns across BTC, ETH, and SOL that asset-specific models trained on single instruments cannot observe, and that proved decisive during the FTX and Terra-Luna contagion episodes.

Table 3: Baseline Comparison—Binary, Seed 42, Stress-Class F1

Fold	Stress%	Persistence	Majority	XGBoost	Lift
1	31.2	0.566	0.000	0.793	+0.227
2	9.9	0.539	0.000	0.896	+0.357
3	11.0	0.517	0.000	0.907	+0.390
4	41.5	0.746	0.000	0.964	+0.218

Lift = XGBoost – Persistence, stress-class F1.

**Figure 1: XGBoost SHAP feature attribution (Fold 4, Pooled). RV and Intensity dominate; Kyle's λ ranks third, consistent with adverse selection amplification during stress.**

Baseline comparison and class prevalence. To confirm the headline F1 reflects genuine predictive skill rather than regime momentum or class imbalance, Table 3 compares XGBoost against a persistence baseline (predict today's label equals yesterday's) and a majority-class baseline. Stress class prevalence ranges from 9.9% (Fold 2) to 41.5% (Fold 4). Despite Fold 4 having the highest stress prevalence, XGBoost achieves a stress-class F1 of 0.9642 against a persistence stress F1 of 0.7461 and a majority-class stress F1 of 0.000—an absolute lift of +0.218 over persistence on the hardest fold. The lift is larger still in Folds 2 and 3, where stress prevalence is under 11% and persistence stress F1 falls to 0.517–0.539. The high weighted F1 is not an artifact of label autocorrelation; XGBoost captures structural microstructure dynamics that a momentum rule cannot recover.

4.2 Feature Attribution and Ablation

SHAP analysis (Fig. 1) consistently identifies RV and Intensity as dominant predictors, with Kyle's λ third. This ranking is theoretically consistent: RV measures price instability, trade intensity captures the urgency of order submission during stress episodes, and Kyle's λ encodes depth deterioration—the core mechanistic drivers of liquidity stress [7]. Surrogate model fidelity (mean 91.8%, range: 83.9%–98.4%) validates these attributions.

Table 4: Lead-Time Analysis—XGBoost Binary, Seed 42, Pooled

Event	Fold	To HMM Onset	To External Ref.
May 2021 Crash	1	persistent [†]	persistent [†]
Terra-Luna Collapse	2	108.3 min	108.3 min
FTX Bankruptcy	3	176.0 min	56.0 min [‡]

[†]Signal present throughout extended 12-hour search window; true onset predates lookback.[‡]External ref: CZ tweet / FTT sell order at 12:00 UTC. HMM onset at 14:00 UTC.

Fractional differencing improves stress-class F1 for linear and tree-based models (LR: +0.046; RF: +0.018). XGBoost shows negligible change, having saturated the signal. CNN-GAF substantially underperforms across all folds (binary stress F1: 0.209 ± 0.156), confirming that domain-informed tabular microstructure features aggregate temporal dynamics more effectively than image-based representations for this problem class: the single-timestep feature engineering already encodes the structure that a convolutional architecture would need to rediscover from raw visual patterns.

Cross-evaluation falsification test. A direct test of whether XGBoost learns the HMM's feature geometry rather than an independent stress signal is to cross-evaluate: train on local HMM labels and test against global labels, and vice versa. When XGBoost is trained and evaluated on the same local label set, it achieves F1-W 0.9917 and stress F1 0.9881—consistent with learning a coherent decision boundary. When the same locally-trained model is evaluated against global labels, weighted F1 collapses to 0.8066 and stress F1 drops to 0.4802. The production model trained on global labels achieves F1-W 0.9801 and stress F1 0.9065. This asymmetry demonstrates that the two label sets encode genuinely different information: a model that has learned one cannot approximate the other. XGBoost is not recovering latent mixture structure from its own inputs—it is learning a stress signal that is specific to the global regime calibration, which the external validation framework independently confirms to be economically meaningful.

4.3 Predictive Lead-Time Analysis

Table 4 reports lead times against two reference definitions: HMM-defined stress onset and externally documented crisis timestamps. For Terra-Luna, the external timestamp aligns with HMM onset (108.3 min); both measure the same structural break. For FTX, the external reference is the publicly documented trigger (CZ tweet and FTT sell order, 12:00 UTC), which precedes HMM-defined onset by two hours. XGBoost issued a sustained warning at 11:04 UTC—56 minutes before the external trigger and 176 minutes before the HMM transition. The 56-minute figure is the more conservative and more honest measure: it confirms genuine advance warning against a publicly observable event rather than merely anticipating an HMM state transition in shared feature space.

The May 2021 event requires separate treatment. The model issued a sustained warning at the start of the 12-hour search window and maintained it continuously; extending the window further would yield larger numbers without additional meaning. This behaviour is consistent with the character of the event: May 2021 was a prolonged regime deterioration, not an acute collapse. The model

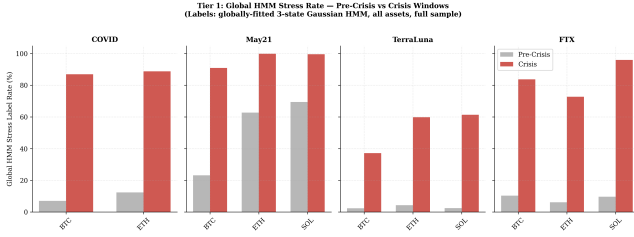


Figure 2: Global HMM stress rate during pre-crisis vs crisis windows. All four events show statistically significant elevation ($p < 0.0001$, proportion z-test). Terra-Luna shows moderate BTC elevation (37.2%), consistent with DeFi-specific contagion.

reflects this distinction—it remained in a sustained elevated state throughout the deterioration rather than producing a single concentrated pre-collapse signal. This sensitivity to crisis *type* is itself an informative result: the same framework that provides a clean 56-minute warning before an acute exchange failure also captures the diffuse multi-day microstructure deterioration that preceded the 2021 crash.

Terra-Luna and FTX show no price drawdown breach within the search window for the 5% rolling threshold. This is consistent with the nature of both events: Terra-Luna was a stablecoin depeg and FTX was a solvency event—structural failures that manifested first in microstructure and order flow before producing sustained price declines in the major pairs. The absence of a price-drawdown reference for these events reinforces rather than undermines the framework’s claim: order flow stress preceded observable price impact.

4.4 External Validity of HMM Stress Labels

4.4.1 Global HMM Stability and Local Instability. The global HMM is near-deterministic: ETH/SOL show 0.00×10^0 parameter difference between independent training runs; BTC shows 2.66×10^{-7} (floating-point noise). This is not a random local optimum—it is the unique convergent solution at this data scale.

Local HMMs fitted using only fold-available training data are systematically miscalibrated. Stress percentages range from **6.0% to 94.3%** for identical assets and periods. SOL Fold 2 assigns 94.3% stress to a 7-month window; BTC Fold 3 assigns 78.4% during the post-FTX stabilisation period—a miscalibration caused by the local model having never observed a recovery regime. This instability is not a data starvation problem: it persists in SOL Fold 2 with 13 months of training data because narrow windows lack cross-cycle regime diversity.

The controlled experiment design is critical: features, architecture, fitting procedure, and state labelling are identical across global and local HMMs. The only variable is the data horizon. When global outperforms local against external judges, the conclusion is isolated: microstructure stress detection requires cross-cycle training breadth.

4.4.2 Tier 1: Known Crisis Validation. Proportion z-tests confirm significant stress elevation during all four documented events (Fig. 2).

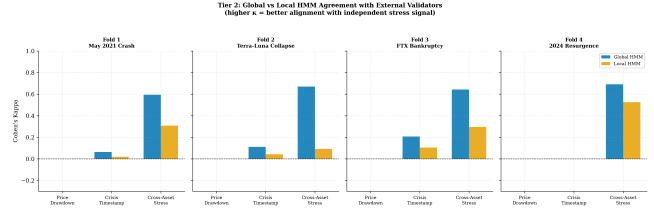


Figure 3: Cohen’s κ between global (blue) and local (orange) HMM labels and three external validators across all folds. Global HMM consistently outperforms local, most decisively on the cross-asset validator where local collapses to slight or near-zero agreement.

Table 5: Tier 2: Cohen’s κ —Cross-Asset Validator (selected)

Fold	Asset	Global κ	Local κ	Global	Local
1	ETHUSDT	0.829	0.334	almost perfect	fair
2	ETHUSDT	0.762	0.052	substantial	slight
2	SOLUSDT	0.778	0.021	substantial	slight
3	BTCUSDT	0.775	0.088	substantial	slight
4	ETHUSDT	0.840	0.621	almost perfect	substantial

Landis & Koch (1977) [15] interpretation.

All z-scores exceed 100 (range: 102.9–333.8, $p < 0.001$). These tests assume approximately independent observations; financial bar data exhibit serial dependence, and crisis windows are persistent by construction, which inflates standard errors relative to i.i.d. baselines. A circular block bootstrap robustness check directly addresses this concern: n_{crisis} -length resamples are drawn from the 30-day pre-crisis baseline using circular block bootstrap (block size = 1 day = 288 bars, 10,000 replicates), constructing a null distribution with the same variance as the observed crisis statistic. This confirms significance for all four events across all assets at $p < 0.0001$, with sensitivity confirmed at 4-day blocks.¹ The results are therefore strongly indicative rather than exact, but robust to serial dependence under all tested block sizes. Cross-asset simultaneous stress confirms systemic contagion: FTX shows 72.1% of bars with all three assets simultaneously stressed; May 2021 shows 100% any-asset simultaneous stress across the full crisis window.

4.4.3 Tier 2: Global vs Local HMM External Validation. Table 5 and Fig. 3 report Cohen’s κ against three external validators. Global HMM outperforms or matches local in every fold-asset combination. The cross-asset validator is most discriminating: global achieves substantial to almost perfect agreement while local collapses to slight or near-zero in multiple folds (e.g. ETH Fold 2: global $\kappa = 0.762$ vs local $\kappa = 0.052$; SOL Fold 2: global 0.778 vs local 0.021). These are the same folds where local HMM assigned 57.5% and 94.3% stress respectively—a miscalibrated HMM cannot co-occur with a sparse external signal.

Crisis timestamp kappa is lower for both variants by design: 3–4 day windows inside 6–7 month test periods guarantee sparse

¹Full bootstrap output including per-asset pre-crisis and crisis stress rates and 95% confidence intervals is available in the supplementary repository at <https://github.com/Goodie-Goody/maic>.

chance-corrected agreement. This is expected and informative—global still consistently outperforms local even on this unfavourable metric.

The results are consistent with global HMM labels capturing genuine economic stress rather than closing the self-referentiality critique entirely. The external validators confirm that global labels behave as stress indicators should: they elevate during documented crises, align with cross-asset contagion, and degrade systematically when the HMM is miscalibrated. Residual circularity through shared volatility clustering cannot be fully eliminated without non-volatility-linked validators such as funding rates or depth data, which we identify as a direction for future work.

Importantly, the external validators confirm label validity but could not replace the microstructure labelling scheme. Crisis timestamps cover only ≈ 12 days of 1,440—insufficient for continuous bar-level labels. Cross-asset stress is circular as a primary labelling mechanism. Price drawdown detects the *outcome* of stress, not its onset. Only microstructure features detect order flow deterioration before price reflects it—the mechanism behind the 56–176 minute lead times against external reference definitions.

4.4.4 Tier 3: Silent Event Discovery. Scanning the full history outside known crisis windows identifies 15 stress episodes exceeding 60 minutes. The most intense is an ETH episode beginning 2024-11-24, lasting approximately 1,471 hours; five of the ten longest episodes identified are ETH events clustered in November–December 2024. Silent events demonstrate the framework’s forward-looking utility beyond named retrospective events.

4.4.5 Robustness to Fully-Causal Relabelling. A stronger test than local Viterbi decoding alone is available: an HMM refit per fold on strictly prior data (as in Section 4.4), decoded with a causal forward filter that conditions each label only on observations up to and including that bar—eliminating look-ahead not just across fold boundaries but within the test window itself. As a control, replacing only the decoding step—holding the same globally-fit model, substituting causal forward-filtering for Viterbi—changes labels minimally (97–98% agreement with the production Viterbi labels; Tier 2 kappa differs by at most 0.03 across all fold-asset-validator combinations). Applying the fully-causal procedure instead and re-computing Tier 2 kappa against the same three external validators, global HMM labels outperform in 29 of 30 fold-asset-validator comparisons. Notably, the fully-causal procedure independently reproduces the exact local-HMM kappa values reported in Table 5 for ETH Fold 2 ($\kappa = 0.052$) and SOL Fold 2 ($\kappa = 0.021$), despite differing methodologically in both fitting and decoding procedure from the local HMM baseline. This convergence, together with the negligible effect of decoding alone, indicates the instability originates specifically in training-window breadth, reinforcing the conclusion of Section 4.4.

4.5 Discussion

XGBoost dominance reflects the effectiveness of domain-informed feature engineering: microstructure features temporally aggregate what sequential architectures would need to re-learn. The cross-evaluation falsification test and the three-tier external validation

framework together establish three conclusions beyond model performance. First, global HMM labels are consistent with genuine economic stress, as confirmed by independent validators. Second, XGBoost performance is not reducible to label geometry recovery: a locally-trained model evaluated against global labels collapses on stress F1 from 0.988 to 0.480, demonstrating that the two label sets encode genuinely different information. Third, cross-cycle training breadth is a critical design requirement: narrow-horizon HMMs do not produce more temporally honest labels—they produce miscalibrated ones.

5 Conclusion

This paper presents a high-frequency early warning system for liquidity stress in Binance spot cryptocurrency markets, shifting the analytical focus from retrospective forensic analysis to prospective detection. By engineering natively stationary microstructure features from publicly accessible trade data, we eliminate the reliance on proprietary order book feeds. Extensive walk-forward evaluation across 18.8 million observations demonstrates that XGBoost achieves a binary weighted F1 of **0.9706**, substantially outperforming a persistence baseline (F1-W 0.7891) and providing 56 minutes of advance warning before the documented FTX market trigger and 108 minutes before the Terra-Luna structural break—measured against externally documented reference definitions rather than HMM-defined onset.

Our three-tier external validation framework establishes that regime labelling requires cross-cycle global calibration to prevent systematic miscalibration—locally-fitted HMMs produce stress percentages ranging from 6.0% to 94.3% for identical assets and periods, while the global HMM achieves substantial to almost perfect Cohen’s κ against genuinely independent external validators. The results are consistent with global HMM labels capturing economically meaningful stress regimes. A cross-evaluation falsification test further confirms that XGBoost performance is not reducible to label geometry recovery: stress F1 collapses from 0.988 to 0.480 when a locally-trained model is evaluated against global labels, demonstrating the two label sets encode genuinely different information. This conclusion is independently reinforced by a fully-causal relabelling procedure—walk-forward HMM refitting combined with causal forward-filter decoding—which outperforms local Viterbi decoding in 29 of 30 fold-asset-validator comparisons against the same external validators. The global HMM is near-deterministic (2.66×10^{-7} maximum parameter difference across independent runs), confirming it represents the unique convergent solution at this data scale, not a random local optimum.

By providing the complete data pipeline and a live Binance API inference system (<https://github.com/Goodie-Goody/maic>), this work bridges the gap between academic market microstructure theory and deployable real-world trading infrastructure. The current framework is intentionally scoped to Binance spot markets for BTC, ETH, and SOL — a deliberate design choice that enables deep longitudinal analysis across 4.2 billion trades and five distinct market regimes, at the cost of cross-venue generalisability. Future work will extend the framework to perpetual funding rates and cross-exchange fragmentation as independent stress signals, investigate non-price-derived supervised targets to further isolate classification performance from

label generator geometry, and explore multimodal architectures incorporating NLP sentiment signals as complementary early warning indicators.

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